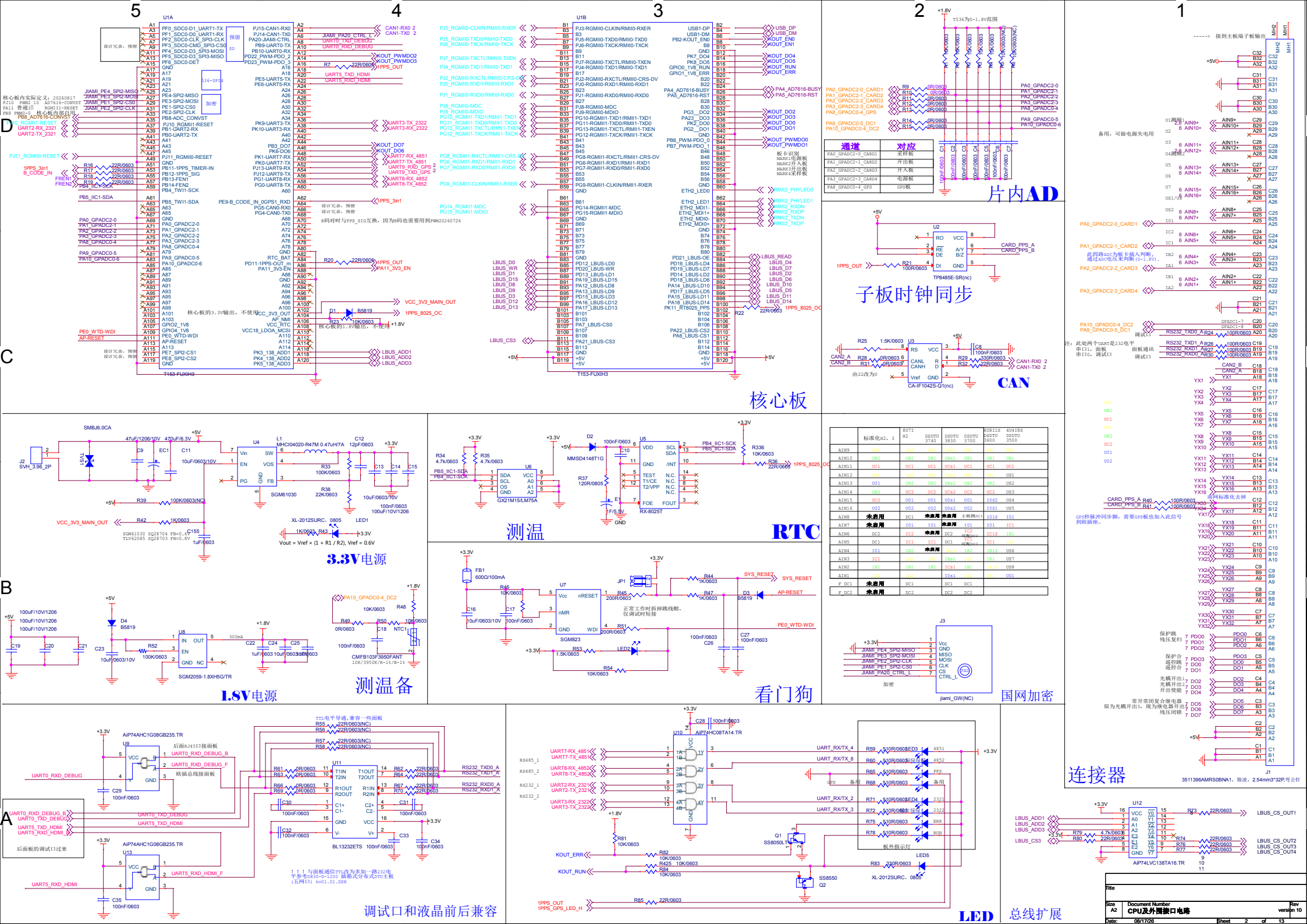
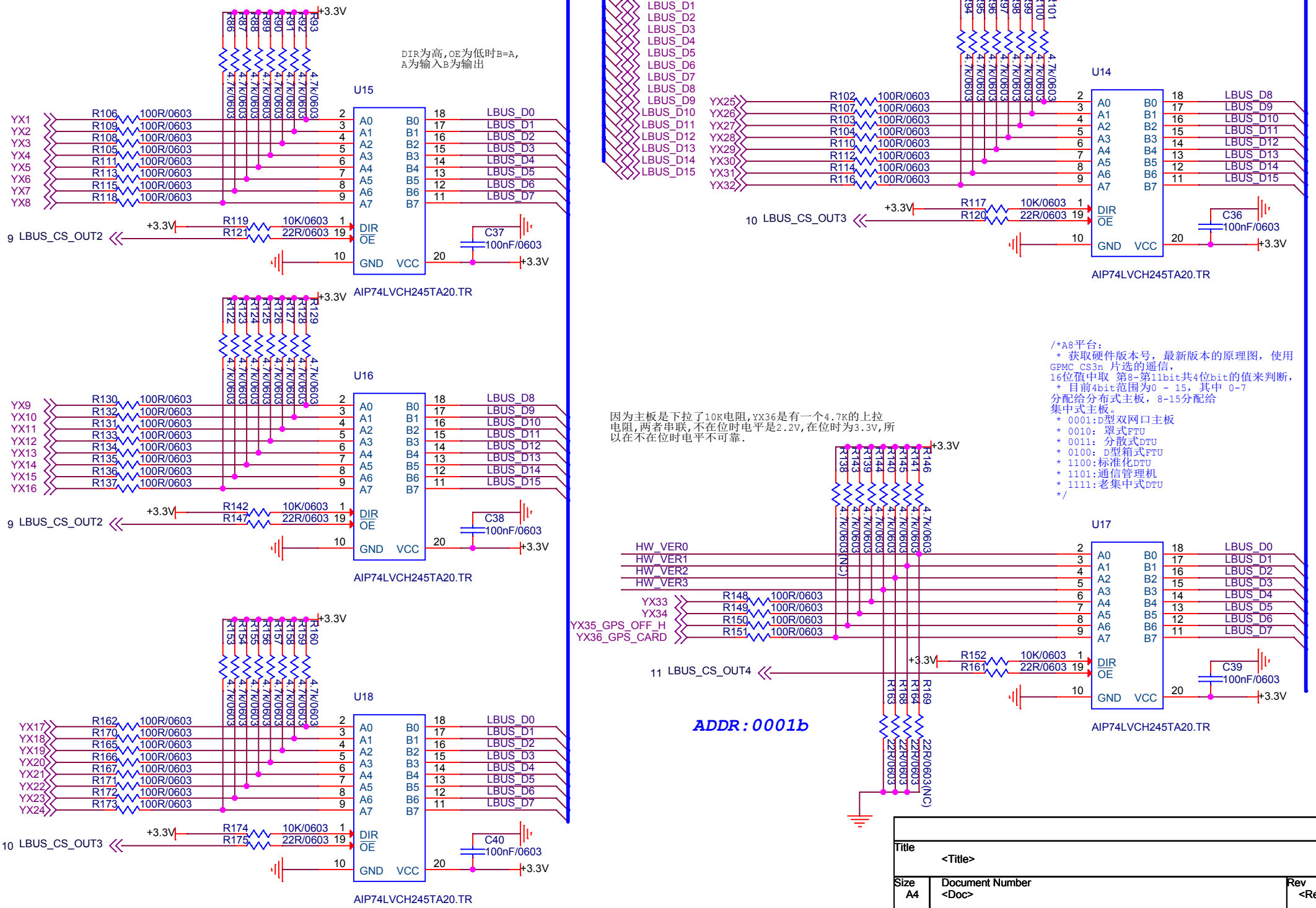
D830-NB-1200

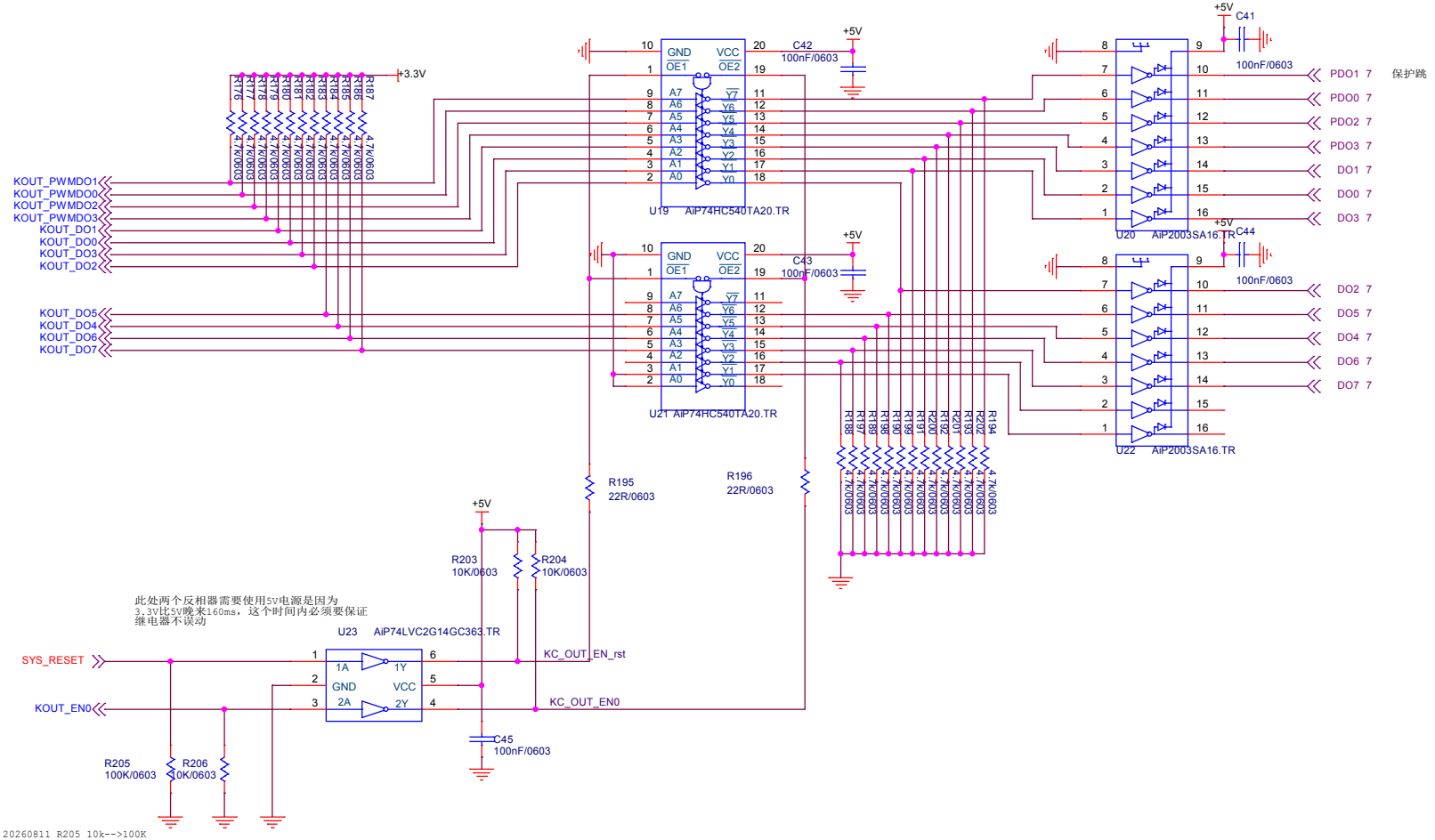
日期：2026年4月

图页	名称	功能描述
1	0-D830-NB-1200-Index	索引
2	1-D830-NB-1200-CPU_POWER.SCH	CPU及外围接口电路
3	2-D830-NB-1200-NET0.SCH	以太网电路-网口0
4	3-D830-NB-1200-NET1.SCH	以太网电路-网口1
5	4-D830-NB-1200-NET2.SCH	以太网电路-网口2
6	5-D830-NB-1200-ADC.SCH	模拟量采集电路
7	7-D830-NB-1200-GPS.SCH	GPS部分电路
8	8-D830-NB-1200-FEN	测频电路
9	9-D830-NB-1200-COM.SCH	串口通讯电路
10	10-D830-NB-1200-YX.SCH	遥信电路
11	11-D830-NB-1200-KC.SCH	开出电路
12	12-D830-NB-1200-杂项	螺丝、MARK等杂项
13	更改说明	版本迭代说明
备注		

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核心板
片内AD
子板时钟同步
CAN
3.3V电源
测温
RTC
测温备
1.8V电源
看门狗
国网加密
连接器
调试口和液晶前后兼容
LED
总线扩展


DIR为高,OE为低时B=A,
A为输入B为输出
U15
U16
U17
U18
U14
AIP74LVCH245TA20.TR
LBUS_CS_OUT2
LBUS_CS_OUT3
LBUS_CS_OUT4
+3.3V
/*A8平台:
* 获取硬件版本号，最新版本的原理图，使用GPMC_CS3n 片选的遥信，
16位值中取 第8-第11bit共4位bit的值来判断，
* 目前4bit范围为0 - 15，其中 0-7分配给分布式主板，8-15分配给集中式主板。
* 0001:D型双网口主板
* 0010: 罩式FTU
* 0011: 分散式DTU
* 0100: D型箱式FTU
* 1100:标准化DTU
* 1101:通信管理机
* 1111:老集中式DTU
*/
因为主板是下拉了10K电阻,YX36是有一个4.7K的上拉电阻,两者串联,不在位时电平是2.2V,在位时为3.3V,所以在不在位时电平不可靠.
ADDR:0001b
Title
<Title>
Size A4
Document Number <Doc>
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M1 M2 M3 M18

MARK MARK MARK MARK

UT-MARK UT-MARK UT-MARK UT-MARK

M4 M5 M6

MARK MARK MARK

UT-MARK UT-MARK UT-MARK

M7 M8 M9

MARK MARK MARK

UT-MARK UT-MARK UT-MARK

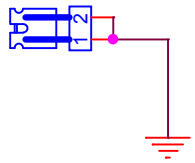
H1

XFCN_PKT-TH_4P

H2

XFCN_PKT-TH_4P

JP2

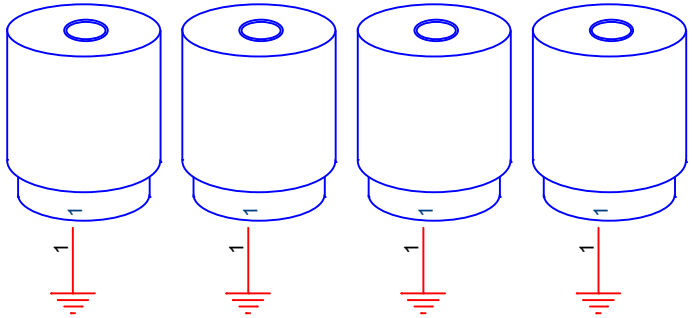


CY1 2.2nF/Y1/7854

CY2 2.2nF/Y1/7854

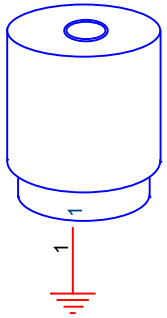
GNDB

M10 STANDE_D3.1_M2 M11 STANDE_D3.1_M2 M12 STANDE_D3.1_M2 M13 STANDE_D3.1_M2



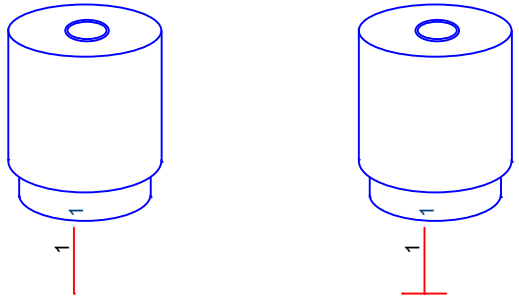
核心板

M14 STANDE_D5.5_M2



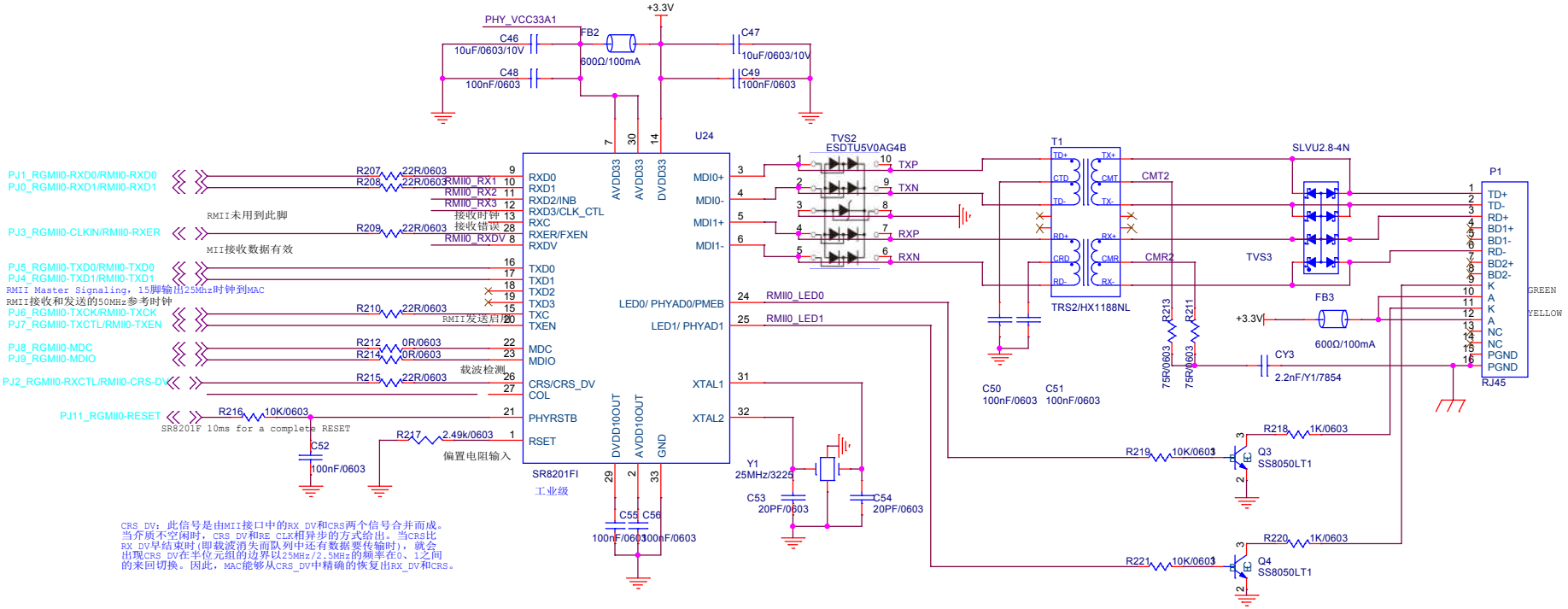
加密

M15 STANDE_D5.5_M2 M16 STANDE_D5.5_M2

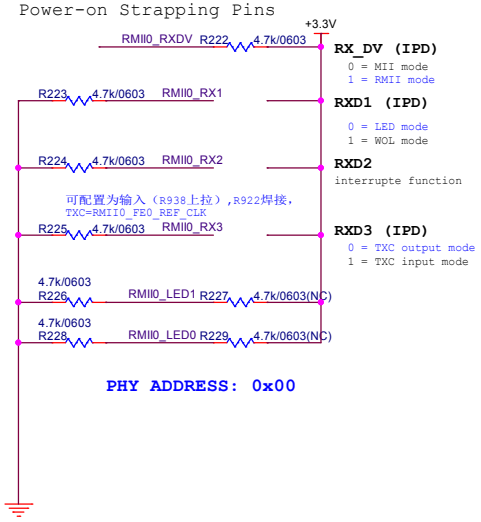


GPS

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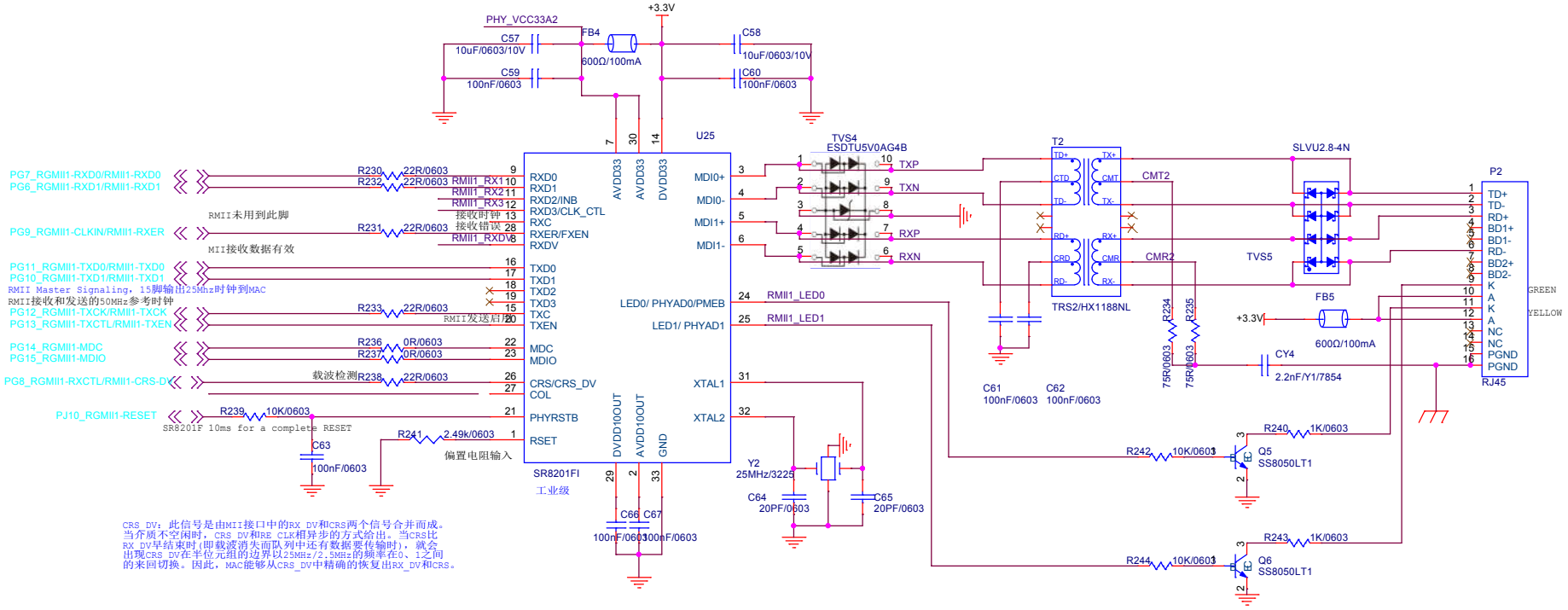
Power-on Strapping Pins



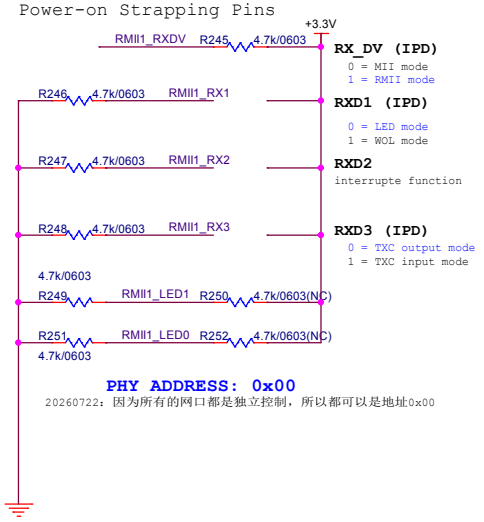
PHY Address	LED1 PIN	LED0 PIN
00	0	0
01	0	1
10	1	0
11	1	1

LED_SEL	00	01	10	11
LED0	ACT ALL	LinkALL ACTALL	Link10 ACTALL	Link10 ACT10
LED1	LINK 100	LINK 100	LINK 100	LINK100 ACT100

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Power-on Strapping Pins



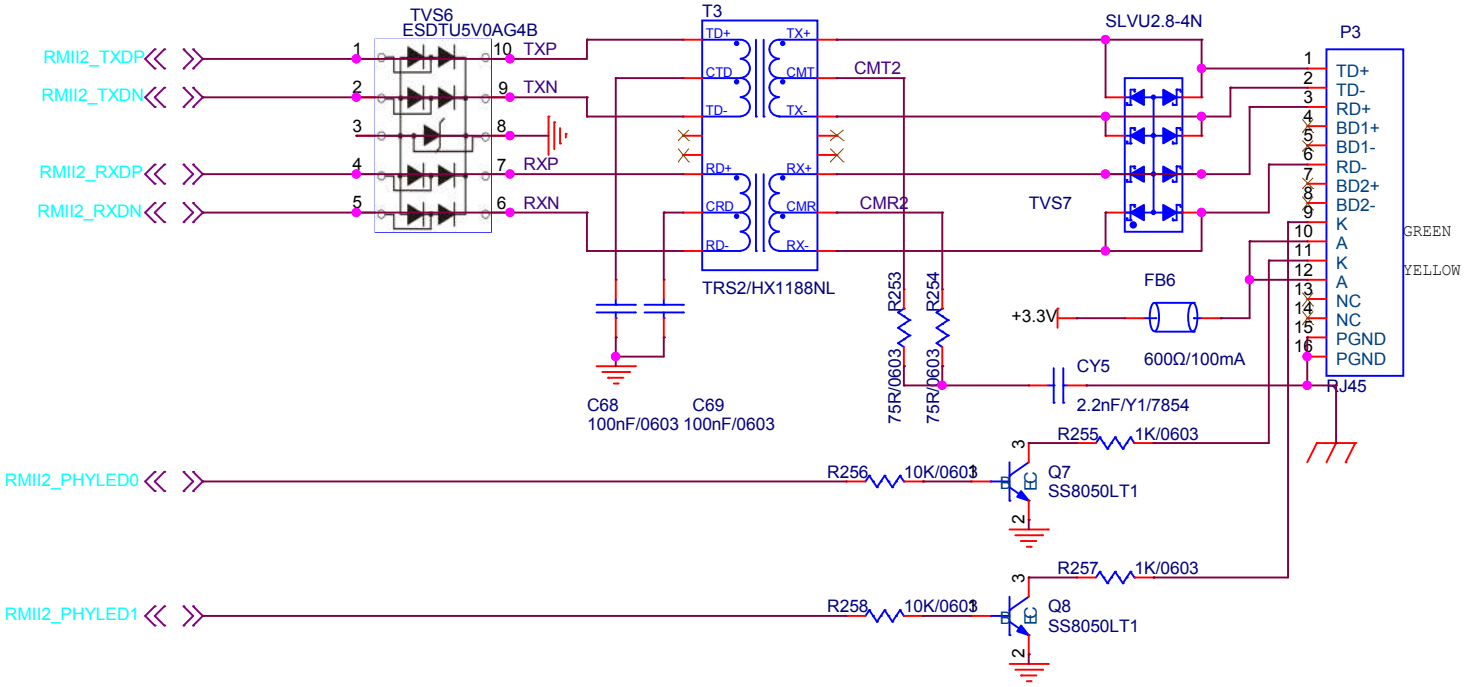
PHY ADDRESS: 0x00

20260722：因为所有的网口都是独立控制，所以都可以是地址0x00

PHY Address	LED1 PIN	LED0 PIN
00	0	0
01	0	1
10	1	0
11	1	1

LED_SEL	00	01	10	11
LED0	ACT ALL	LinkALL ACTALL	Link10 ACTALL	Link10 ACT10
LED1	LINK 100	LINK 100	LINK 100	LINK100 ACT100

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PHY ADDRESS: 0x11

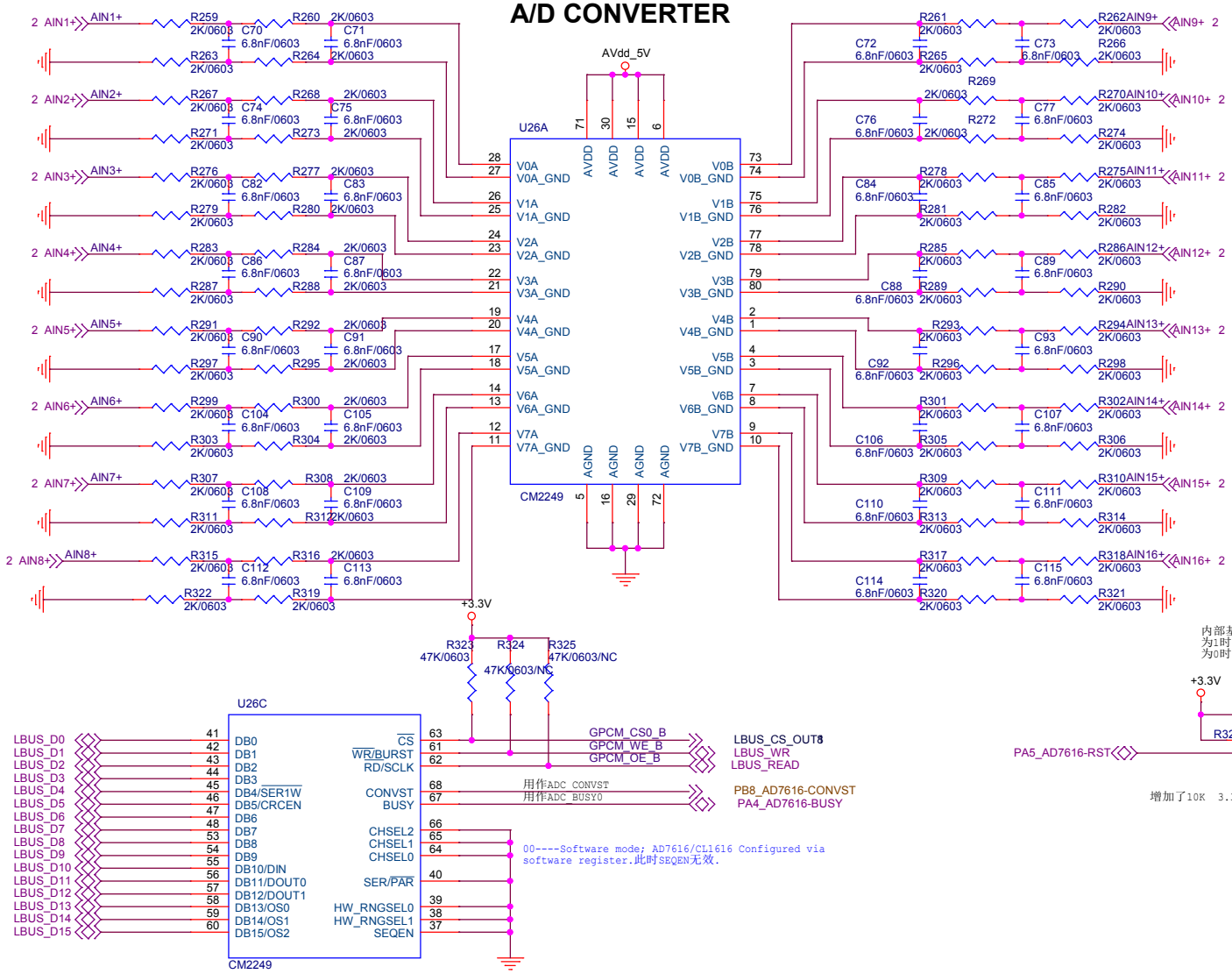
PHY Address	LED1 PIN	LED0 PIN
00	0	0
01	0	1
10	1	0
11	1	1

LED_SEL	00	01	10	11
LED0	ACT ALL	LinkALL ACTALL	Link10 ACTALL	Link10 ACT10
LED1	LINK 100	LINK 100	LINK 100	LINK100 ACT100

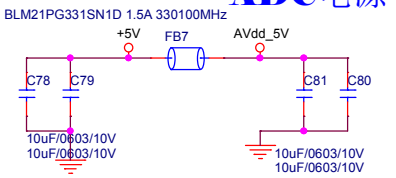
PHY芯片位于核心板上

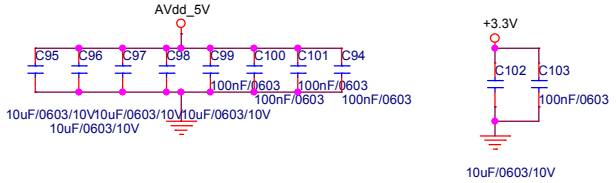
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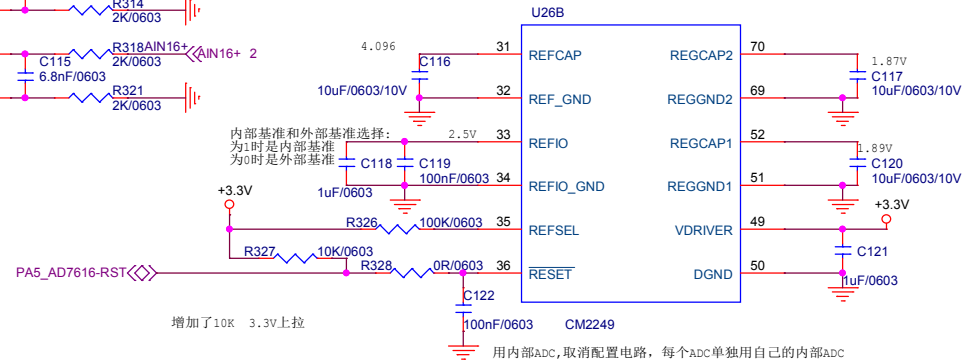
A/D CONVERTER



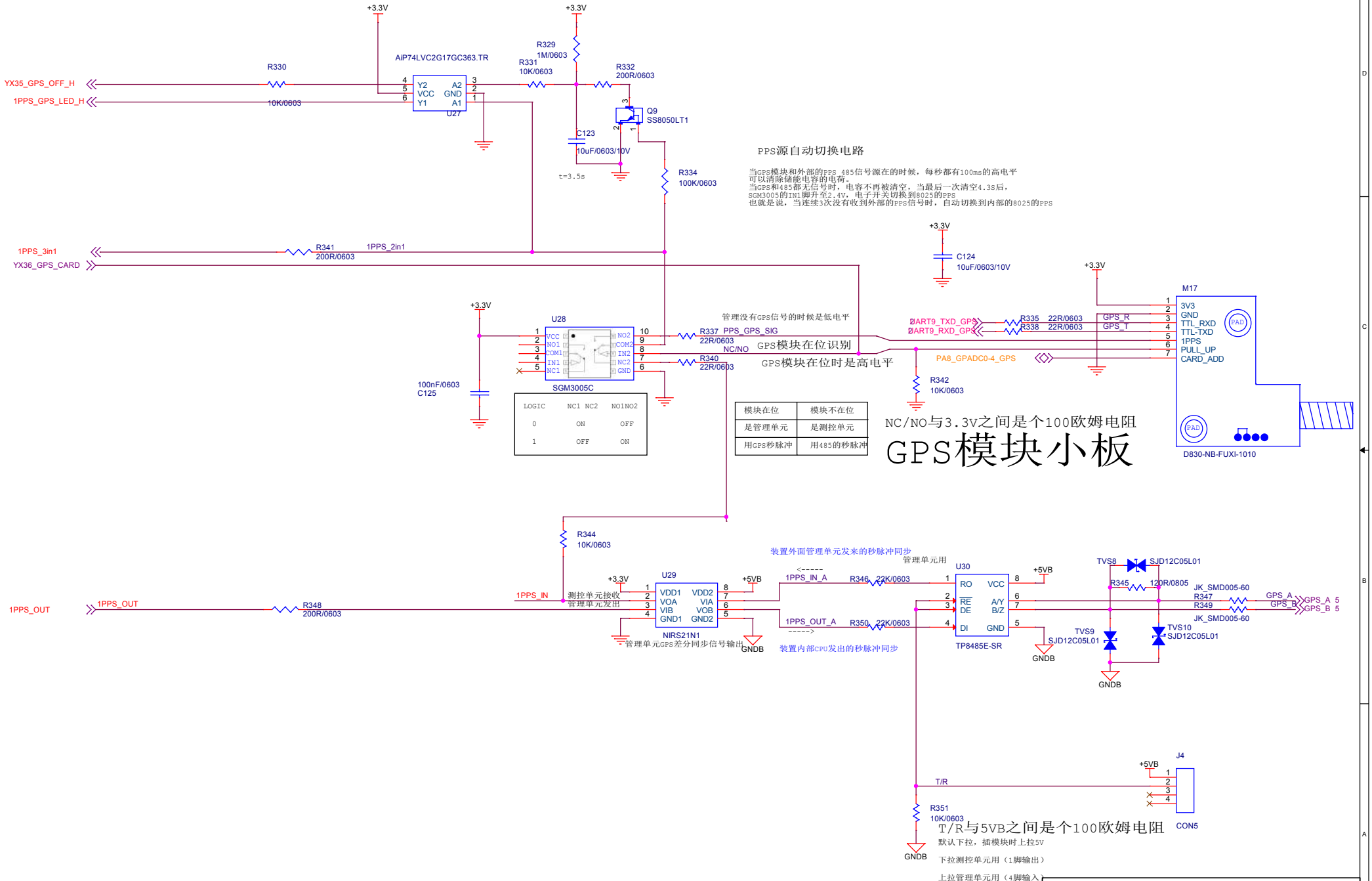
ADC电源



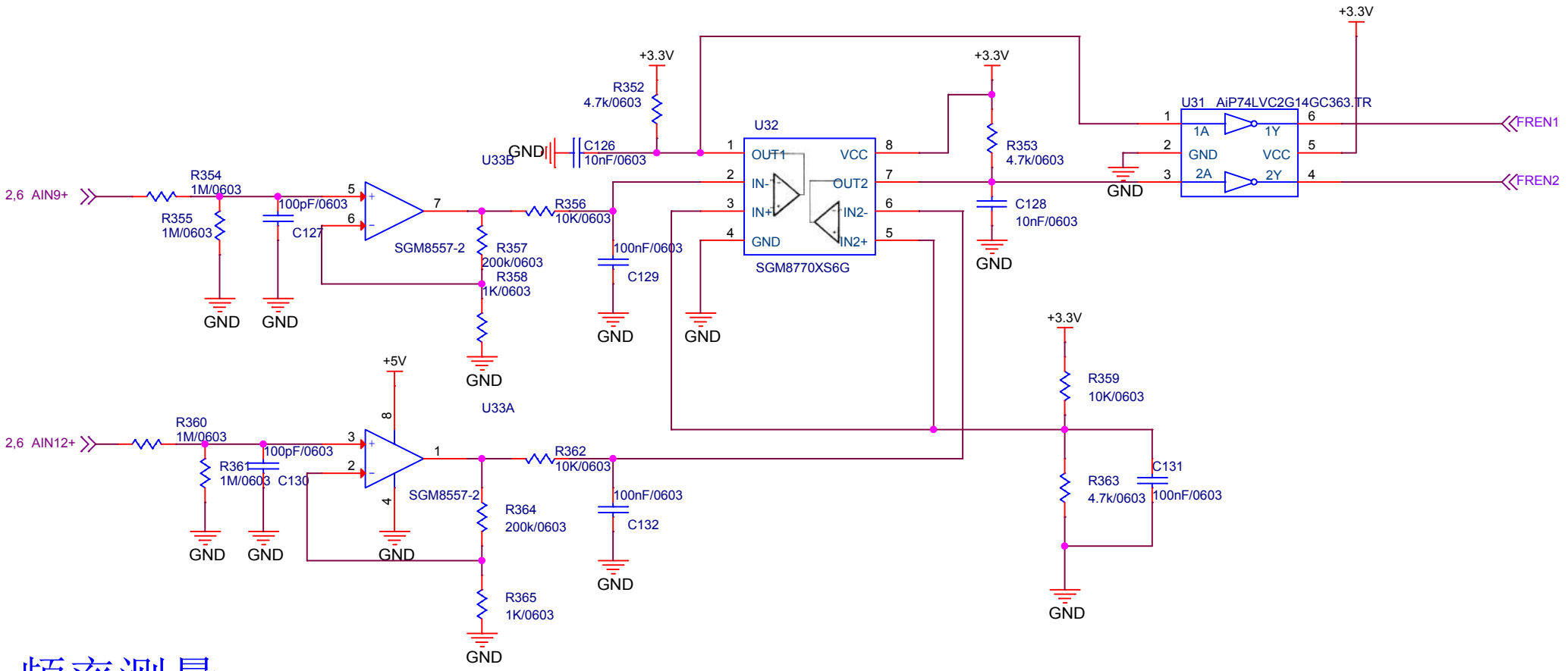




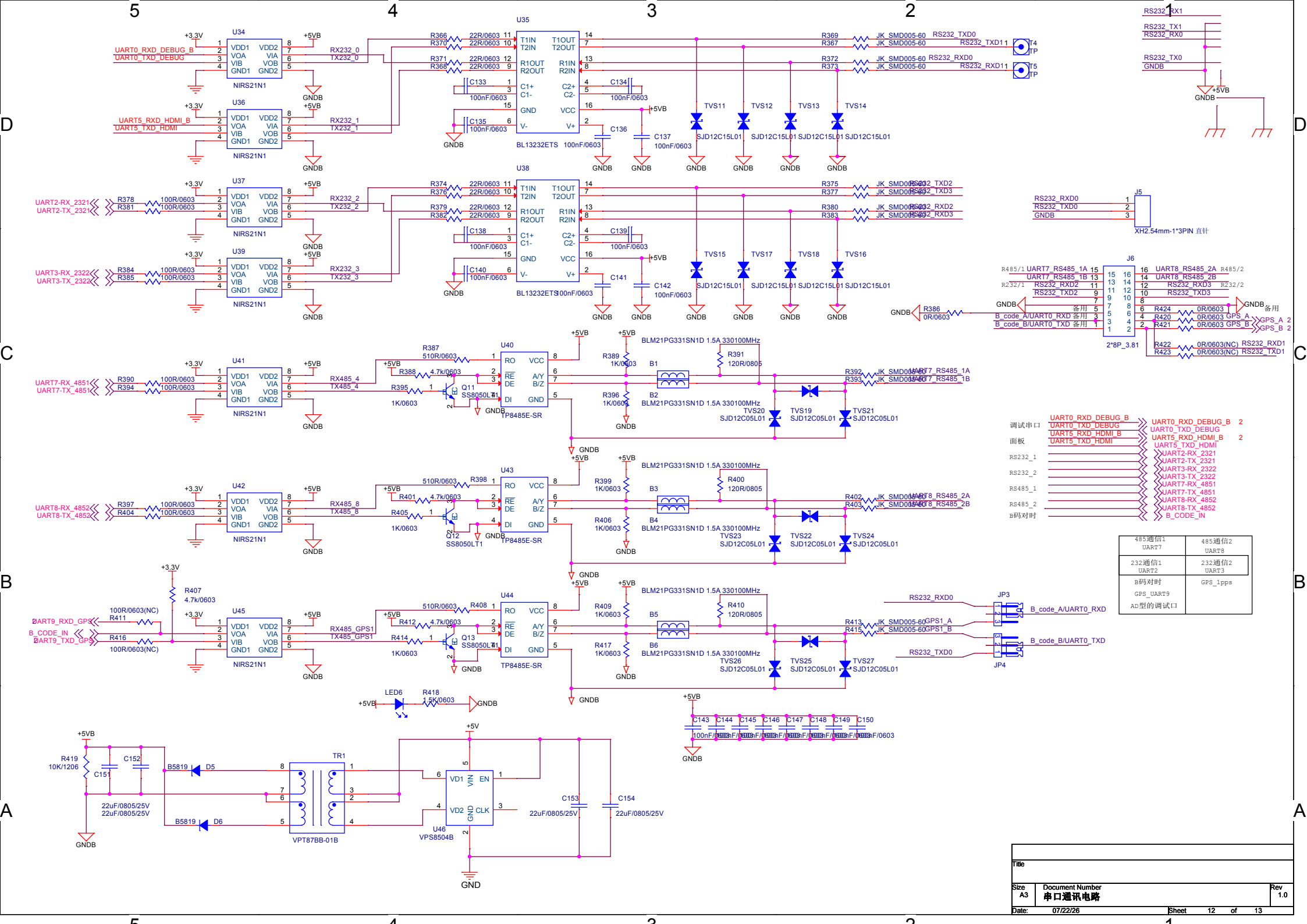
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U34
U35
U36
U37
U38
U39
U40
U41
U42
U43
U44
U45
U46
NIRS21N1
BL13232ETS
TP8485E-SR
VPS8504B
VPT87BB-01B
UART0_RXD_DEBUG_B
UART0_TXD_DEBUG
UART5_RXD_HDMI_B
UART5_TXD_HDMI
UART2-RX_2321
UART2-TX_2321
UART3-RX_2322
UART3-TX_2322
UART7-RX_4851
UART7-TX_4851
UART8-RX_4852
UART8-TX_4852
UART9_RXD_GPS
UART9_TXD_GPS
B_CODE_IN
RS232_TXD0
RS232_TXD1
RS232_RXD0
RS232_RXD1
RS232_TXD2
RS232_TXD3
RS232_RXD2
RS232_RXD3
GPS1_A
GPS1_B
B_code_A/UART0_RXD
B_code_B/UART0_TXD
GNDB
GND
+5VB
+5V
+3.3V
TVS11
TVS12
TVS13
TVS14
TVS15
TVS16
TVS17
TVS18
TVS19
TVS20
TVS21
TVS22
TVS23
TVS24
TVS25
TVS26
TVS27
SJD12C15L01
SJD12C05L01
J5
J6
JP3
JP4
XH2.54mm-1*3PIN 直针
2*8P_3.81
调试串口
面板
485通信1 UART7
485通信2 UART8
232通信1 UART2
232通信2 UART3
B码对时 GPS_UART9
AD型的调试口
GPS_1pps
Title
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20260413
新设计 刘鹏飞
在南网标准化分布式DTUH3版本基础上进行设计

20260722
HV01.02 刘鹏飞
准备小批量备货，更改了上一版的部分问题

Title		
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